YABIBAL ESHETIE MOLLA

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PROFILE

Software engineer with experience building production software, AI-enabled applications, and model-evaluation workflows, including RLHF-related tasks, AI benchmarking, machine-learning challenge creation, retrieval-augmented generation (RAG), and Python development. Currently strengthening foundations in machine learning, probability, statistics, and mathematics, with deep learning as the next study focus. Interested in understanding how machine-learning systems are developed, evaluated, and made reliable.

EDUCATION

Addis Ababa University, Addis Ababa, Ethiopia 2020 – 2025
B.Sc. in Information Science, School of Information Science

- Passed the national exit exam.
- **Relevant coursework:** Mathematics for Natural Sciences, Applied Mathematics I, Discrete Mathematics and Combinatorics, Introductory Statistics, Data Structures and Algorithms, Introduction to Data Science and Analytics, Introduction to Artificial Intelligence, Knowledge Management, Research Methods in Information Systems, Industrial Project I and II.

A2SV (Africa to Silicon Valley) 2024 – 2025

- Completed a 16-month intensive Data Structures and Algorithms program covering dynamic programming, graphs, trees, sorting, and algorithmic problem-solving using Python and C++.

Additional Preparation Ongoing

- Currently studying algebra, calculus, probability and statistics, and machine learning; preparing to begin deep learning. Completed coursework/certificates in NumPy and pandas.

AI / ML EXPERIENCE

Turing — *RLHF, Benchmarking and Python Development*

- Worked on tasks involving reinforcement learning from human feedback (RLHF), AI benchmarking, and Python development.

Deccan AI — *Benchmark Engineer*

- Contributed to AI benchmarking work focused on evaluating model performance.

Shipid AI — *Machine Learning Challenge Creator*

- Created machine-learning challenges to assess technical problem-solving and machine-learning capabilities.

SELECTED PROJECTS

Aegis Agent *Python, FastAPI, RAG, LangGraph-style orchestration, Supabase pgvector* 2026

- Built an agentic RAG backend powering the chat assistant on yabibal.site, with intent routing, confidence-gated answers, and human handoff.
- Used BGE-small embeddings and Supabase pgvector for retrieval; the system abstains when retrieval confidence is below a configured threshold.

RAG Evaluation Framework *Python, MLflow*

- Developed an evaluation workflow for RAG applications with automated test generation, multi-dimensional scoring, hallucination checks, golden datasets, and regression tracking.
- Designed deterministic pass/fail evaluation and MLflow-based experiment tracking.

Compliance Navigator *FastAPI, Next.js, PostgreSQL/pgvector, Neo4j* In progress

- Building a graph-based decision-support platform for Ethiopian business compliance, mapping regulatory and public-service requirements into step-by-step roadmaps.
- Integrated an English/Amharic AI assistant using retrieval over legal and service-catalog sources, with references to supporting material.
- Uses a Neo4j knowledge graph and PostgreSQL/pgvector for structured relationships and retrieval.

BillShare *Flutter, FastAPI, Supabase* 2026 – Present

- Building an expense-sharing platform with itemized purchases, custom splits, live-derived balances, settlement recommendations, and payment requests.
- Added spending analytics, real-time room chat, and an assistant that retrieves information from verified balances and expense history.
- Covered core purchase, balance, ledger, and insights logic with 26 unit tests.

PROFESSIONAL EXPERIENCE

LohnPulse — *Senior Full-Stack Developer* 2026 – Present

- Building a production payroll SaaS for tax advisors with role-based access control and row-level security.
- Implemented Next.js 15 partial pre-rendering, edge delivery of the static shell, and streamed user data through React Server Components.
- Added per-tenant cache tagging, JWT verification, and tenant context in edge middleware.
- Maintain 194+ Playwright end-to-end, API, and UI tests that gate production releases.

Orizon Technology — *Senior Lead, AI Engineering* 2025 – 2026

- Led architecture, development, and deployment of AI-enabled applications for enterprise clients; mentored junior engineers.
- Designed human-in-the-loop validation, monitoring, and recovery workflows for production LLM deployments.
- Contributed to responsible-AI practices, including governance guidance and model-evaluation processes.

Pest Leads Pro — *Full-Stack Engineer and Automation Lead* 2025

- Built a lead-generation platform with SEO landing pages, acquisition funnels, an A/B-configurable AI chat widget, and lead-quality scoring.
- Developed an automated agent workflow for lead scoring, vendor matching, and credit management.
- Built automated digital-PR workflows and scheduled background agents for SEO, performance, audit, and reporting.

TapToSign — *Full-Stack Developer* 2024 – 2025

- Built REST APIs in Go with PostgreSQL and Redis for a digital-signature platform serving international clients.
- Developed a Preact and TypeScript application; containerized services with Docker and established CI/CD using GitHub Actions.

DOT Ethiopia — *MERN Developer and Backend Development Lead* 2024 – 2025

- Led backend architecture for three MERN applications using TypeScript and Node.js.
- Implemented JWT-cookie authentication, XSS/CSRF protections, Redis caching, and CI/CD workflows.

chrisFitApp (via Hubstaff) — *MERN Stack Developer* 2024 – 2025

- Built a real-estate platform using MongoDB, Express, TypeScript, Node.js, Next.js, and Tailwind CSS.

Aldia — *Nuxt.js Front-End and QA Developer* 2024

- Built Cypress end-to-end test suites and integrated them into CI/CD using component-driven development and Storybook.

Upwork — *Freelance Full-Stack Developer*

- Delivered software and automation projects for international clients; Top Rated with a 100% Job Success score.

LEADERSHIP & COMMUNITY

n8n — *Official Ambassador, Addis Ababa* 2026 – Present

- Support local developer learning through workshops and meetups focused on production automation.

Lumen Lab — *Chief Technology Officer* 2025

TECHNICAL SKILLS

Programming:	Python, TypeScript, JavaScript, C++, Go, SQL
AI / ML:	RLHF task experience, AI benchmarking, machine-learning challenge design, RAG, LLM evaluation, Hugging Face, MLflow, pandas, NumPy
Backend:	FastAPI, Node.js, NestJS, Express, Go (Gin), REST, GraphQL
Frontend / Mobile:	React, Next.js, Flutter, Preact, Nuxt.js, Tailwind CSS
Data / Infrastructure:	PostgreSQL, Supabase pgvector, MongoDB, Redis, DynamoDB, Neo4j, Docker, AWS, GitHub Actions
Testing / Automation:	PyTest, Playwright, Cypress, Jest, n8n, Make